

B.TECH
(SEM III) THEORY EXAMINATION 2022-23
MATERIAL & ENERGY BALANCE

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 10 = 20**

- a) What is limiting and excess reactant?
- b) Define conversion and degree of completion
- c) State Dalton's law.
- d) Convert 500 in³/day to cm³/min.
- e) What is dry bulb and wet bulb temperatures?
- f) Write down different forms of conversion of temperature.
- g) Convert 20,000 ppm into wt. %.
- h) What do you mean by molarity?
- i) Convert 2 km/hr to miles/sec
- j) Define heat capacity

SECTION-B**2. Attempt any three of the following:****10 x 3 = 30**

- a) For the given reaction: $\text{Sb}_2\text{S}_3 + 3\text{Fe} \rightarrow 2\text{Sb} + 3\text{FeS}$, suppose that 0.600 kg of stibnite (sb) and 0.250 kg of iron (Fe) are heated together to give 0.200 kg of Sb metal. Determine: the limiting reactant, % of excess reactant, degree of completion, % conversion and the yield. (mol. Wt. of Sb = 122)
- b) The heat transfer coefficient in a heat exchanger is 1200 kcal/(hr)(m²)(°C). Convert this value to Btu./(hr)(ft²)(°F)
- c) Explain the following terms: (i) Heat of formation (ii) Heat of combustion (iii) Hess Law
- d) A natural gas has the following composition by volume CH₄=83.5%, C₂H₆=12.5%, and rest N₂. Calculate Composition in mole%, Composition in Wt%, Avg. molecular Wt, Density of gas at standard conditions.
- e) Calculate the heat of formation of ethane gas at 298.15 K (25°C) from its element using hess's law.

Data:

Heat of formation of CO₂(g) = -393.51 kJ/moleHeat of formation of H₂O (l) = -285.83 kJ/mole

Heat of combustion of ethane gas at 298.15 K (25°) = -1560.69 kJ/mole

SECTION C**3. Attempt any one part of the following:****10 x 1 = 10**

- a) Cracked gas from a petroleum refinery has the following composition by volume: methane: 50%, ethane: 15%, ethylene : 25%, propane:7%, propylene:8%, n-butane: 5%.Find (i) average molar mass of the gas mixture, (ii) the composition by mass
- b) Oxidation of ethylene to produce ethylene oxide is given by reaction.
 $\text{C}_2\text{H}_4 + 1/2 \text{O}_2 \rightarrow \text{C}_2\text{H}_4\text{O}$
 If air is used 20% in excess of that theoretically, Calculate the quantity of ethylene required for the production of 100 kmol ethylene oxide.

4. Attempt any *one* part of the following:

10 x 1 = 10

- a) A gaseous mixture (F) consists of 16 mol% CS₂ and 84 mol% air are fed to the absorption column at a rate of 1000 lb mole/hr. Most of the CS₂ input are absorbed by liquid benzene (L) which is fed to the top of the column. 1 % of benzene input are evaporated and out with the exit gas stream which consists of 96 mol% air, 2 mol% CS₂ and 2 mol% benzene. The product liquid stream (P) consists of benzene and CS₂. Calculate the mole flow rates of (G), (L) and (P) and the compositions.
- b) The spent acid from a nitrating process contain 43.4% H₂SO₄, 11.3% HNO₃ and 45.3% water by weight. This acid is to be strengthened by the addition of conc. Sulphuric acid containing 98% H₂SO₄ and conc. Nitric acid containing 90% HNO₃. The strengthened mixed acid is to contain 60% H₂SO₄ and 32% HNO₃. Calculate the quantities of spent and concentrated acids that should be mixed together to yield 1500 kg of the desired mixed acid

5. Attempt any *one* part of the following:

10 x 1 = 10

- a) Along with the mathematical expression write short notes on (i) Absolute humidity, (ii) Molal humidity, (iii) Relative humidity, (iv) Percentage humidity, (v) Saturation humidity, (vi) Humid volume, (vi) Humid heat.
- b) A limestone consisting 90% CaCO₃ and 10% MgCO₃ is decomposed by heating, find how many grams of CaO can be made from one ton of this limestone and how many grams of CO₂ can be recovered per kg of limestone.

6. Attempt any *one* part of the following:

10 x 1 = 10

- a) Explain the recycling, purging and bypassing operations with the help of diagram.
- b) Write short notes on the following:
 - (i) Molality and normality
 - (ii) Amagat's law

7. Attempt any *one* part of the following:

10 x 1 = 10

- a) A tank holds 100 gallon of water -salt solution in which 4 pounds of salt is dissolved. Water runs into the tank at the rate of 5gallen/min and the salt solution overflows at the same rate. If the mixing in the tank is adequate to keep the concentration of salt in the tank uniform at all times, how much salt is in the tank at the end of 50 min? Assume that density of the salt solution is essentially the same as that of water.
- b) Pure methane is heated from 303K to 523K at atmospheric pressure. Calculate the heat added per kmole methane using C_pdata given below.
 $C_p = a + bT + cT^2 + dT^3$.
Data for methane:
 $a = 19.2494$, $b \times 10^3 = 52.1135$, $c \times 10^6 = 11.973$, $d \times 10^9 = -11.3173$